Chapter 1. Probability Distributions

Axioms, conditional probability, random variables, moments, generating functions, inequalities

Contents

1.1 Random experiments and probability **2**

1.2 Conditional probability and independence **3**

1.3 Random variables and their distributions **4**

1.4 Expectation, variance and moments **5**

1.5 Distribution functions and generating functions **5**

1.6 Inequalities **7**

1.1 Random experiments and probability

Motivation. Statistics infers properties of a population from random samples, so it needs a precise model of randomness. Kolmogorov's three axioms are that model; every rule of probability in these notes is derived from them.

A **random experiment** has outcomes that cannot be predicted in advance; its **sample space** S is the set of all outcomes, and **events** are subsets of S. When all N outcomes of a finite S are equally likely, $P(A) = \#A/N$ (classical probability); in general a probability is any assignment obeying Kolmogorov's axioms:

(A1) $P(A) \geq 0$ for every event A;

(A2) $P(S) = 1$;

(A3) (countable additivity) if $A_1, A_2, \ldots$ are pairwise disjoint, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

Theorem 1.1 (Basic properties)

(a) $P(\emptyset) = 0$, and finite additivity holds.

(b) $P(A^c) = 1 - P(A)$, and $0 \leq P(A) \leq 1$.

(c) (Monotonicity) $A \subseteq B \Rightarrow P(A) \leq P(B)$, and $P(B \setminus A) = P(B) - P(A)$.

(d) (Addition rule) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; more generally, inclusion–exclusion

$$P\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \cdots + (-1)^{n-1} P(A_1 \cap \cdots \cap A_n).$$

(e) (Countable subadditivity) $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

Proof. (a) Apply (A3) to $S \cup \emptyset \cup \emptyset \cup \cdots$: $1 = 1 + \sum P(\emptyset)$ forces $P(\emptyset) = 0$; padding a finite union with empty sets gives finite additivity.

(b) $S = A \cup A^c$ disjointly.

(c) $B = A \cup (B \setminus A)$ disjointly.

(d) $A \cup B = A \cup (B \setminus (A \cap B))$ disjointly and $P(B \setminus (A \cap B)) = P(B) - P(A \cap B)$ by (c); the general formula follows by induction on n, applying the two-set rule to $(A_1 \cup \cdots \cup A_{n-1}) \cup A_n$.

(e) Put $B_1 = A_1$, $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$; these are disjoint with the same union and $B_n \subseteq A_n$. ∎

Theorem 1.2 (Continuity of probability)

If $A_1 \subseteq A_2 \subseteq \cdots$ then $P(\bigcup_n A_n) = \lim_n P(A_n)$; if $B_1 \supseteq B_2 \supseteq \cdots$ then $P(\bigcap_n B_n) = \lim_n P(B_n)$.

Proof. With $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1}$ (disjoint, same union),

$$P\Big(\bigcup_n A_n\Big) = \sum_k P(C_k) = \lim_n \sum_{k \leq n} P(C_k) = \lim_n P(A_n).$$

For decreasing B_n, apply this to $A_n = B_n^c$ and take complements. ∎

Example 1.1

Let $P((a,b]) = b - a$ for subintervals of $[0,1]$ (the uniform probability). For a point b, $\{b\} = \bigcap_n (b - \frac{1}{n}, b]$, so $P(\{b\}) = \lim_n \frac{1}{n} = 0$: every single point has probability zero, although some point always occurs. Likewise $P([a,b]) = P((a,b)) = b - a$.

Example 1.2

Roll three fair dice. $P(\text{at least one 6}) = 1 - P(\text{no 6}) = 1 - (5/6)^3 = 91/216 \approx 0.42$. Inclusion–exclusion gives the same: $3 \cdot \frac{1}{6} - 3 \cdot \frac{1}{36} + \frac{1}{216} = \frac{91}{216}$.

1.2 Conditional probability and independence

Motivation. Data change our probabilities: conditional probability restricts attention to outcomes consistent with what has been observed. Bayes's rule reverses the conditioning, and independence is the special case where observing one event tells nothing about another.

Definition (Conditional probability)

For $P(A) > 0$, $P(B \mid A) = P(A \cap B)/P(A)$. As a function of B it satisfies (A1)–(A3), so it is a probability on the reduced sample space A.

Theorem 1.3

(a) (Multiplication) $P(A \cap B) = P(B \mid A)P(A)$, and $P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$.

(b) (Total probability) If $A_1, A_2, \ldots$ partition S with $P(A_i) > 0$, then $P(B) = \sum_i P(B \mid A_i)P(A_i)$.

(c) (Bayes) In the same setting, with $P(B) > 0$,

$$P(A_j \mid B) = \frac{P(B \mid A_j)P(A_j)}{\sum_i P(B \mid A_i)P(A_i)}.$$

Proof. (a) is the definition, iterated.

(b) $B = \bigcup_i (B \cap A_i)$ disjointly; apply (A3) and (a).

(c) $P(A_j \mid B) = P(B \cap A_j)/P(B)$; expand the numerator by (a) and the denominator by (b). ■

$P(A_j)$ are **prior** and $P(A_j \mid B)$ **posterior** probabilities.

Example 1.3 (Diagnostic test)

A disease has prevalence 1%; a test detects it with probability 0.95 and gives false positives with probability 0.05. Then

$$P(\text{disease} \mid +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0095}{0.059} \approx 0.16.$$

A positive result raises the probability only from 1% to 16% because healthy people vastly outnumber sick ones.

Example 1.4 (Three machines)

Machines A, B, C make $50\%, 30\%, 20\%$ of output with defect rates $2\%, 3\%, 5\%$. Then $P(\text{defective}) = 0.5(0.02) + 0.3(0.03) + 0.2(0.05) = 0.029$ and $P(\text{C} \mid \text{defective}) = 0.010/0.029 \approx 0.34$.

Definition (Independence)

A, B are independent if $P(A \cap B) = P(A)P(B)$ (equivalently $P(B \mid A) = P(B)$ when $P(A) > 0$). $A_1, \ldots, A_n$ are (mutually) independent if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for every subcollection.

If A, B are independent, so are A, B^c (and A^c, B^c): $P(A \cap B^c) = P(A) - P(A)P(B) = P(A)P(B^c)$. Pairwise independence does not imply mutual independence:

Example 1.5

Toss two fair coins: $A = \{\text{first H}\}$, $B = \{\text{second H}\}$, $C = \{\text{the two agree}\}$. Each has probability $\frac{1}{2}$ and each pair intersects with probability $\frac{1}{4}$, but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$.

Example 1.6 (Parallel system)

A device works if at least one of three independent components works; each fails with probability 0.1. $P(\text{works}) = 1 - 0.1^3 = 0.999$.

1.3 Random variables and their distributions

A **random variable** X is a real-valued function on S; events such as $\{X \le x\} = \{s : X(s) \le x\}$ inherit probabilities, and the probabilities $P(X \in A)$ form the **distribution** of X.

- **Discrete** X takes values $x_1, x_2, \ldots$; its probability mass function (pmf) $f(x) = P(X = x)$ satisfies $f \ge 0$, $f = 0$ off $\{x_k\}$, $\sum_x f(x) = 1$, $P(a \le X \le b) = \sum_{a \le x \le b} f(x)$.
- **Continuous** X has a probability density function (pdf) $f \ge 0$ with $\int f = 1$ and $P(a \le X \le b) = \int_a^b f(x)\,dx$. Then $P(X = a) = 0$ and $P(a \le X \le a + \Delta) \approx f(a)\Delta$, so $f(a)$ is a probability per unit length, not a probability.

We write $X \sim f$; the indicator $I_A(x)$ gives compact formulas such as $f(x) = 2x I_{[0,1]}(x)$.

Example 1.7

Toss a fair coin until the first head; the number of tosses X has $f(k) = (1/2)^k$, $k = 1, 2, \ldots$, and $\sum_k f(k) = 1$. Two tosses give $Y =$ number of heads with $f_Y(0) = \frac{1}{4}$, $f_Y(1) = \frac{1}{2}$, $f_Y(2) = \frac{1}{4}$.

Example 1.8

If $f(x) = cx(2 - x)$ on $[0, 2]$, then $\int_0^2 x(2 - x)\,dx = \frac{4}{3}$, so $c = \frac{3}{4}$ and $P(X \le \frac{1}{2}) = \frac{3}{4}\int_0^{1/2}(2x - x^2)\,dx = \frac{3}{4}(\frac{1}{4} - \frac{1}{24}) = \frac{5}{32}$.

1.4 Expectation, variance and moments

Motivation. A distribution is a whole function, but we usually summarize it by a few numbers: its center (mean) and spread (variance). Higher moments describe shape, and expectations of functions of X are the building blocks of estimators.

Definition (Expectation)

$E[g(X)] = \sum_x g(x)f(x)$ or $\int g(x)f(x)\,dx$, defined when the sum or integral converges absolutely. $\mu = E(X)$ is the mean (the distribution's center of mass), $\operatorname{Var}(X) = E[(X-\mu)^2]$ the variance and $\operatorname{sd}(X) = \sqrt{\operatorname{Var} X}$ the standard deviation.

Theorem 1.4 (Properties of expectation)

(a) $E(aX+b) = aE(X)+b$.
(b) $E[c_1 g_1(X) + c_2 g_2(X)] = c_1 E g_1(X) + c_2 E g_2(X)$.
(c) $g_1 \le g_2 \Rightarrow E g_1(X) \le E g_2(X)$.

Proof. Linearity and monotonicity of sums and integrals. ■

Theorem 1.5 (Variance)

(a) $\operatorname{Var}(aX+b) = a^2 \operatorname{Var} X$.
(b) $\operatorname{Var} X = E(X^2) - (EX)^2$.

Proof. (a) $aX + b - E(aX+b) = a(X-\mu)$, so $\operatorname{Var}(aX+b) = E[a^2(X-\mu)^2]$.
(b) $E(X-\mu)^2 = E(X^2) - 2\mu EX + \mu^2 = E(X^2) - \mu^2$. ■

The **standardized** variable $Z = (X-\mu)/\sigma$ has $EZ = 0$, $\operatorname{Var} Z = 1$. A mean need not exist: for $f(k) = \frac{1}{k(k+1)}$, $k \ge 1$, $\sum k f(k) = \sum \frac{1}{k+1} = \infty$.

Example 1.9

For the uniform density on $[0,1]$, $\mu = \frac{1}{2}$ and $\sigma^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. For the triangular density $f(x) = 4x$ on $[0, \frac{1}{2}]$ and $4 - 4x$ on $[\frac{1}{2}, 1]$ (same mean), $EX^2 = \frac{7}{24}$ and $\sigma^2 = \frac{7}{24} - \frac{1}{4} = \frac{1}{24}$: the triangle is half as spread.

Example 1.10

With two tosses and a prize of X^2 (in thousands of won) for X heads, $E(X^2) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = 1.5$, so the expected prize is 1,500 won.

1.5 Distribution functions and generating functions

The **cumulative distribution function** (cdf) is $F(x) = P(X \le x)$. For discrete X it is a step function with jumps $f(x_k)$ at the x_k; for continuous X, $F(x) = \int_{-\infty}^{x} f$ and $F'(x) = f(x)$ where f is continuous.

Theorem 1.6 (Properties of a cdf)

(a) F is nondecreasing.
(b) $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$.
(c) F is right-continuous.
(d) $F(a-) = P(X < a)$, hence $F(a) - F(a-) = P(X = a)$.

Proof. (a) $\{X \le x_1\} \subseteq \{X \le x_2\}$.

(b) $\{X \le -k\} \downarrow \emptyset$ and $\{X \le k\} \uparrow S$; apply Theorem 1.2.

(c) $\{X \le x + \frac{1}{n}\} \downarrow \{X \le x\}$.

(d) $\{X \le a - \frac{1}{n}\} \uparrow \{X < a\}$. ■

Conversely every function with (a)–(c) is the cdf of some random variable.

Example 1.11 (A mixed distribution)

A commuter flips a coin: on heads they wait for a shuttle that comes every 10 minutes, so the trip takes $10 + U$ minutes with $U \sim U(0, 10)$; on tails they walk, which takes exactly 15 minutes. The trip time X has

$$F(x) = \tfrac{1}{2}P(10 + U \le x) + \tfrac{1}{2}I(x \ge 15) = \begin{cases} 0, & x < 10, \\ (x-10)/20, & 10 \le x < 15, \\ x/20, & 15 \le x < 20, \\ 1, & x \ge 20, \end{cases}$$

with a jump of $\frac{1}{2}$ at 15: neither discrete nor continuous.

Probability generating function. For X with values in $\{0, 1, 2, \dots\}$, $G(s) = E(s^X) = \sum_x s^x f(x)$ for $|s| < 1$. Since a power series determines its coefficients, $f(k) = G^{(k)}(0)/k!$ and G determines the distribution.

Moment generating function. If $E(e^{tX}) < \infty$ for all t in some $(-h, h)$, $h > 0$, the mgf is $M(t) = E(e^{tX})$; $m_k = E(X^k)$ is the kth moment.

Theorem 1.7 (Properties of the mgf)

If M exists on $(-h, h)$:
(a) all moments exist, $E(X^k) = M^{(k)}(0)$ and $M(t) = \sum_{k\ge0} \frac{E(X^k)}{k!} t^k$ near 0;
(b) if $M_X = M_Y$ on a neighborhood of 0, then X and Y have the same distribution.

Proof of (a). Fix $|t| < h$. Since $|tx|^k/k! \le e^{|tx|} \le e^{tx} + e^{-tx}$, $|t|^k E|X|^k/k! \le M(t) + M(-t) < \infty$, so all moments exist. By Taylor's theorem with integral remainder,

$$\left| e^{tx} - \sum_{k=0}^{n-1} \frac{(tx)^k}{k!} \right| = \left| \frac{(tx)^n}{(n-1)!} \int_0^1 (1-u)^{n-1} e^{utx}\, du \right| \le \frac{|tx|^n}{n!} e^{|tx|}.$$

Since $\frac{|tx|^n}{n!} = 2^{-n}\frac{|2tx|^n}{n!} \le 2^{-n}e^{2|tx|}$, the right side is at most $2^{-n}e^{3|tx|} \le 2^{-n}(e^{3tx} + e^{-3tx})$. Taking expectations, for $|3t| < h$,

$$\left| M(t) - \sum_{k<n} \frac{E(X^k)}{k!} t^k \right| \le 2^{-n}\{M(3t) + M(-3t)\} \to 0.$$

So M equals its power series near 0; differentiating term by term gives $M^{(k)}(0) = E(X^k)$.

(b) is proved with characteristic functions and is beyond these notes. ■

Existence of all moments does not imply existence of the mgf (the lognormal distribution is a counterexample).

Example 1.12

For the coin-until-head variable of Example 1.7: $G(s) = \sum_k (s/2)^k = \frac{s}{2-s}$, and $M(t) = G(e^t) = \frac{e^t/2}{1-e^t/2}$ for $t < \log 2$. Then $M'(0) = 2$: on average two tosses.

Example 1.13 (Standard exponential)

For $f(x) = e^{-x}$, $x > 0$: $M(t) = \int_0^\infty e^{-(1-t)x}\,dx = (1-t)^{-1} = \sum_k t^k$ for $|t| < 1$, so $E(X^k) = k!$. (Also directly by repeated integration by parts.)

Example 1.14 (Recovering a distribution from moments)

Suppose the mgf exists and $E(X^k) = 3^k k!$. Then $M(t) = \sum_k (3t)^k = (1-3t)^{-1}$ for $|t| < \frac{1}{3}$. Since $\frac{1}{3}e^{-x/3}$ $(x > 0)$ has exactly this mgf, uniqueness shows X has this density.

Cumulants. $C(t) = \log M(t) = \sum_{r\geq 1} c_r \frac{t^r}{r!}$ defines the cumulants c_r. Expanding $\log(1+A) = A - \frac{A^2}{2} + \frac{A^3}{3} - \cdots$ with $A = m_1 t + m_2 \frac{t^2}{2!} + \cdots$ and collecting powers of t gives $c_1 = m_1 = \mu$ and $c_2 = m_2 - m_1^2 = \sigma^2$. For the standard exponential, $C(t) = -\log(1-t) = \sum_r t^r/r$, so $c_r = (r-1)!$; for Example 1.14, $c_r = (r-1)!3^r$.

1.6 Inequalities

Motivation. In statistics we rarely know a distribution exactly, yet we need to bound probabilities and expectations. Markov, Chebyshev and Jensen give such bounds from moments alone and are used in every consistency proof.

Theorem 1.8 (Jensen)

If ϕ is convex on an interval I containing the values of X, and EX exists, then $\phi(EX) \leq E\phi(X)$.

Proof for twice differentiable ϕ. Let $\mu = EX \in I$. Taylor's theorem gives $\phi(x) = \phi(\mu) + \phi'(\mu)(x-\mu) + \frac{1}{2}\phi''(\xi)(x-\mu)^2$ with ξ between x and μ; since $\phi'' \geq 0$, $\phi(x) \geq \phi(\mu) + \phi'(\mu)(x-\mu)$ for all $x \in I$. Take expectations. (For general convex ϕ use a supporting line at μ.) ■

E.g. $(EX)^2 \leq E(X^2)$ and $E\log X \leq \log EX$ for $X > 0$.

Theorem 1.9 (Lyapunov)

If $E|X|^s < \infty$ and $0 < r < s$, then $(E|X|^r)^{1/r} \leq (E|X|^s)^{1/s}$.

Proof. For $p = s/r > 1$, $\phi(y) = y^p I(y \geq 0)$ is convex. $Y = |X|^r$ is integrable, since $|X|^r \leq I(|X| \leq 1) + |X|^s I(|X| > 1) \leq 1 + |X|^s$. Jensen gives $(EY)^p \leq E(Y^p) = E|X|^s$. ■

In particular, if the sth moment exists, all lower moments exist.

Theorem 1.10 (Markov and Chebyshev)

(a) If $E|Z|^r < \infty$ $(r > 0)$, then $P(|Z| \geq k) \leq E|Z|^r/k^r$ for every $k > 0$.
(b) If $\operatorname{Var} X < \infty$, then $P(|X - EX| \geq k) \leq \operatorname{Var} X/k^2$.

Proof. (a) $P(|Z| \geq k) = E\, I(|Z| \geq k)$, and pointwise $I(|Z| \geq k) \leq (|Z|/k)^r I(|Z|/k \geq 1) \leq |Z|^r/k^r$.
(b) Take $Z = X - EX$ and $r = 2$. ■

Example 1.15

Whatever the distribution, $P(|X - \mu| \geq 6\sigma) \leq 1/36 \approx 0.028$; for a normal population the exact value is about 2×10^{-9}. Chebyshev is universal but crude.

Theorem 1.11 (Zero variance)

If $\operatorname{Var} X = 0$ then $P(X = \mu) = 1$.

Proof. $\{|X - \mu| > 0\} = \bigcup_n \{|X - \mu| \geq \frac{1}{n}\}$, and by Chebyshev each event has probability at most $\operatorname{Var} X \cdot n^2 = 0$. Countable subadditivity gives $P(|X - \mu| > 0) = 0$. ■